



Quality of life and mental health in adolescents with idiopathic scoliosis undergoing brace treatment: a systematic review and pooled analysis of quantitative studies

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Abstract

Introduction Adolescent idiopathic scoliosis (AIS) is the most common paediatric spinal deformity. While bracing limits curve progression, its psychological and social effects remain unclear. This review and pooled analysis evaluate bracing's impact on quality of life (QoL) and mental health in AIS.

Methods Searches were conducted in MEDLINE, EMBASE and PsycINFO (Dec 2024). Quantitative studies involving AIS patients undergoing brace treatment were included. Primary outcomes were QoL scores and reported mental health impact, assessed using validated tools. Risk of bias was assessed. A narrative synthesis was completed, with a pooled analysis to show effect sizes.

Results The searches identified 78 studies. A narrative review of the QoL questionnaires in cross-sectional studies, cohort studies and comparisons of brace treatment versus observation in AIS patients found mild and often non-statistically significant reduction in psychological well-being, more social difficulties, impaired physical functioning, more discomfort and lower self-image. These changes were often transient, improving or stabilising during bracing and mostly resolving after brace treatment. Longer brace prescriptions (more hours per day) and bulkier brace designs were negative factors. Family support, brace adherence and bracing success seemed to have a positive effect. Studies comparing brace treatment with surgery found surgically treated patients had mild and often non-statistically significantly improved QoL, self-image and psychological well-being but reduced satisfaction. Most studies showed a moderate risk of bias. The review included a pooled analysis of 51 studies using SRS-22 ($n = 4193$) with an overall, mean SRS-22 score during bracing of 4.05 (95% CI: 3.96, 4.13). Comparing bracing with observation (8 studies, $n = 2374$) shows a small QoL difference favouring no bracing (pooled effect size: -0.11 ; 95% CI: $-0.21, 0.00$), with extreme heterogeneity ($I^2 = 83\%$).

Conclusion Bracing generally resulted in lower scores for psychological issues, physical and social function and self-image compared to no treatment. Overall pooled differences were small and braced patients retained good QoL. Long-term data suggest improvement after brace treatment completion. Given high heterogeneity and risk of bias, high-quality research is needed to better evaluate psychosocial outcomes during and after AIS bracing including confounding factors and factors that may improve QoL.

Level of evidence IV (systematic review of level IV studies).

Keywords Scoliosis · Spine · Adolescent · Idiopathic · Brace · Surgery · Quality of life

Introduction

Adolescent idiopathic scoliosis (AIS) is a structural spinal deformity characterised by a lateral curvature of the spine and vertebral rotation, typically presenting during rapid

growth [1, 2]. AIS affects 1–4% of adolescents and is more common in females [3]. It can cause physical symptoms such as pain and discomfort and have an impact on psychosocial health [4, 5]. The management of AIS depends on the severity of the curve. Curves under 20 degrees are often clinically observed and not treated. The management of moderate curves with remaining growth potential of 20–40 degrees is primarily bracing, however, larger curves exceeding 45–50

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degrees are often referred for surgery [4, 6]. Non-operative management is recommended to minimise curve progression and reduce the need for surgery [7].

The efficacy of bracing is clearly established, but evidence is limited on its broader implications and effects on self-image and health-related quality of life (HRQoL): Physical functioning, mental and emotional well-being, social functioning and pain [8]. It is imperative to evaluate the complete impact of brace treatment by analysing the influence of bracing-associated psychological burdens such as social functioning, anxiety, self-image, on treatment adherence and experience [9].

There has been an effort in recent years to consider patient-reported outcome measures (PROMs) to assess the multidimensional impact of scoliosis and its management [10, 11]. Questionnaires such as the Scoliosis Research Society-22 revised (SRS-22r) are being widely used for assessing domains such as function, pain, self-image, mental health and satisfaction with scoliosis treatment [12, 13]. Particular effects such as psychosocial strain are present in varying degrees in populations treated with a brace when measured using instruments such as Brace Questionnaire (BrQ) and the Bad Sobernheim Stress Questionnaire (BSSQ) [14, 15].

There is a clear gap in the literature on the overall impact of bracing due to the influence of multiple factors such as follow-up period, brace types and bracing regimens causing high heterogeneity in outcome measures [16, 17]. To explore these inconsistencies in reported outcomes, we conducted a systematic review to synthesise existing evidence on mental health and QoL outcomes associated with bracing in AIS. Our primary objective was to identify the extent to which brace treatment impacts patient-reported outcomes and also to explore sources of heterogeneity.

Methods

This systematic review was conducted in accordance with PRISMA guidelines [18] and was registered on PROSPERO (ID: CRD42025650522).

Eligibility criteria

Articles were included if they reported on patients diagnosed with AIS, starting brace treatment aged 10–18 years. Bracing included participants undergoing full or partial wear with a rigid or soft form of brace therapy. Studies with no comparator, or any reported comparator, were included. Studies must have reported on quantitative data on at least one of QoL, function, pain or mental health.

Studies were excluded if they focussed only on physical outcomes. They were also excluded if they had mixed (treatment and/or condition) populations without clear

stratification of results specifically on AIS and/or bracing, or if they involved use of the Milwaukee brace as this is considered obsolete. Likewise, qualitative studies, articles not available in English, conference publications, case reports and reviews were excluded.

Search strategy

Literature searches were conducted through Ovid (MEDLINE, EMBASE and PsycINFO) and the Cochrane Library, from inception to 31st December 2024. The search strategy was developed using a combination of keywords and MeSH terms. Searches were iteratively adapted to better refine the results as well as a review of reference lists of included articles. The full search strategy can be found in Online Appendix 1. A review of grey literature was not done.

Study selection

Screening was carried out independently, in parallel, by two reviewers, with discussion including a third reviewer to reach consensus where needed. Full texts were retrieved for articles included, which were independently screened by two reviewers and any conflicts resolved by a third.

Data extraction

Data extraction was performed by two reviewers individually and a 10% sample checked independently by a third reviewer to confirm accuracy. Data were extracted on demographic information (title, year of publication, authors, study design, study duration, number of patients, location, patient age range, female:male ratio), bracing information (curve type, brace type, duration, hours of use), intervention, questionnaires used (including timepoint) and questionnaire outcomes (including individual domains).

Risk of bias

Risk of bias assessment was conducted using the MINORS tool [19] for observational studies, the Cochrane Risk of Bias tool [20] for RCTs and COSMIN for validation studies [21]. The overall risk classification of studies was done by aggregating the domain judgements using the frequency of high-risk judgements vs. adequate/low-risk judgements across domains.

Data synthesis

The study design classification was determined by when questionnaires were administered, rather than by the overall study design or the authors' stated design. Studies were classified as cross-sectional if questionnaires were completed at

a single timepoint, or if a second timepoint shortly after was included solely for assessing reliability or validity. Studies with more than one timepoint were classified as cohort studies, unless they were randomised controlled trials (RCTs).

Summary statistics were used to describe studies quantitatively. Braces were reviewed for each study and grouped as was felt clinically appropriate. Results were synthesised and presented in narrative format for a qualitative overview of the findings, broken down into the following: psychological well-being; social functioning and peer relationships; and physical well-being and body perception.

Additionally, quantitative data were considered by summarising a subset of studies with sufficient SRS-22 outcome data during bracing as a single-arm, this was completed using the meta package in R (Version 4.4.1). To provide a comprehensive overview of the range of scores, all available data were included regardless of study design, comparator groups or the specific timepoints measured during bracing. Individual manuscripts were listed multiple times if they contained distinct, non-overlapping groups suitable for inclusion. Summary analyses were also completed for studies that compared bracing to no treatment and bracing to surgery using the SRS-22 instrument. Forest plots were created to visualise variability in the study results along with the heterogeneity (using the I^2 statistic) of the studies. Mean differences were calculated for exploratory purposes.

Results

Study selection

The database searches produced 1403 records. 386 duplicates were removed, leaving 1017 screened. Full texts of the resulting 238 studies were reviewed; 78 papers were included in the final review (Fig. 1).

Characteristics of all studies

Individual study characteristics can be found in Table 1, with more detail in Online Appendix 2. Studies were distributed across 24 years from 1999 to 2023, with 32 studies (41%) between 2021 and 2024. The period from 2006 to 2020 shows consistent but lower activity, with two to four studies each year. Only three studies were published between 1999 and 2005. Studies spread across several continents: 35 in Europe, 32 in Asia, eight in North America, two in Africa and one in South America. China was the most common location ($n = 11$). None were based in the United Kingdom.

The research heavily favoured observational designs ($n = 71$, 91% combined). Cross-sectional observational studies (in terms of the questionnaire data) were dominant, comprising 51 studies (65%). Another 20 (26%) were

observational cohort studies, 14 of which were prospective and six retrospective. RCTs accounted for just seven studies (9%).

As this was part of our study inclusion criteria, all patients were diagnosed with AIS. 27 studies applied the SRS bracing criteria when defining their study population; 51 studies did not explicitly report adhering to these criteria [99]. The studies either specified that all curve types were treated or did not specify which curve types were treated ($n = 31$). None of the studies specified that they excluded any curve type. Twenty-three studies did not specify the brace type used. Of those that specified, included braces were heterogeneous (Milwaukee braces were excluded). Most commonly used were the Boston brace ($n = 22$), Cheneau brace ($n = 20$) and 'TLSO brace' ($n = 13$). Sixteen other braces were used by just 1 or 2 studies (see Table 1).

The sample sizes ranged dramatically from 17 to 1,802 participants. Studies with 51–100 participants were most common (27 studies, 35%), followed closely by those with 1–50 participants (24 studies, 31%). Fifteen were medium-sized, with 101–200 participants (19%). Only six studies (8%) had 201–400 participants, and another six had over 500 participants. Overall, there was a strong female predominance in samples, consistent with the literature [100, 101]. Of the 77 that specified, 15 studies included only female participants. Eleven were highly female-dominated ($> 10:1$ ratio) and 42 had high ratios (3:1 to 10:1). Eight had less extreme ratios of under 3:1.

Studies could be categorised into four groups: (i) longitudinal with no comparator group ($n = 31$) [22, 23, 26, 34, 35, 37, 43, 44, 48, 49, 51–53, 55, 59–61, 64, 72, 77–79, 81, 83–85, 91, 92, 94, 95, 98]; (ii) a comparison between two or more different brace types ($n = 10$) [27, 30, 32, 33, 41, 50, 66, 70, 82, 88]; (iii) a comparison between bracing and other treatments i.e. surgery, exercise or no treatment i.e. observation ($n = 29$) [24, 28, 29, 31, 38, 39, 42, 45, 46, 54, 56–58, 62, 65, 67, 68, 73–76, 80, 86, 89, 90, 93, 96, 97, 102]; and (iv) a comparison between bracing and another factor (including regimens and compliance) ($n = 8$) [25, 36, 40, 47, 63, 69, 71, 87].

Outcome measures

The SRS-22 measure was the most commonly used, present in 68% of the included studies. Following this are the BrQ (22%) and the BSSQ-brace (18%) questionnaires. Table 2 shows the breakdown of outcomes.

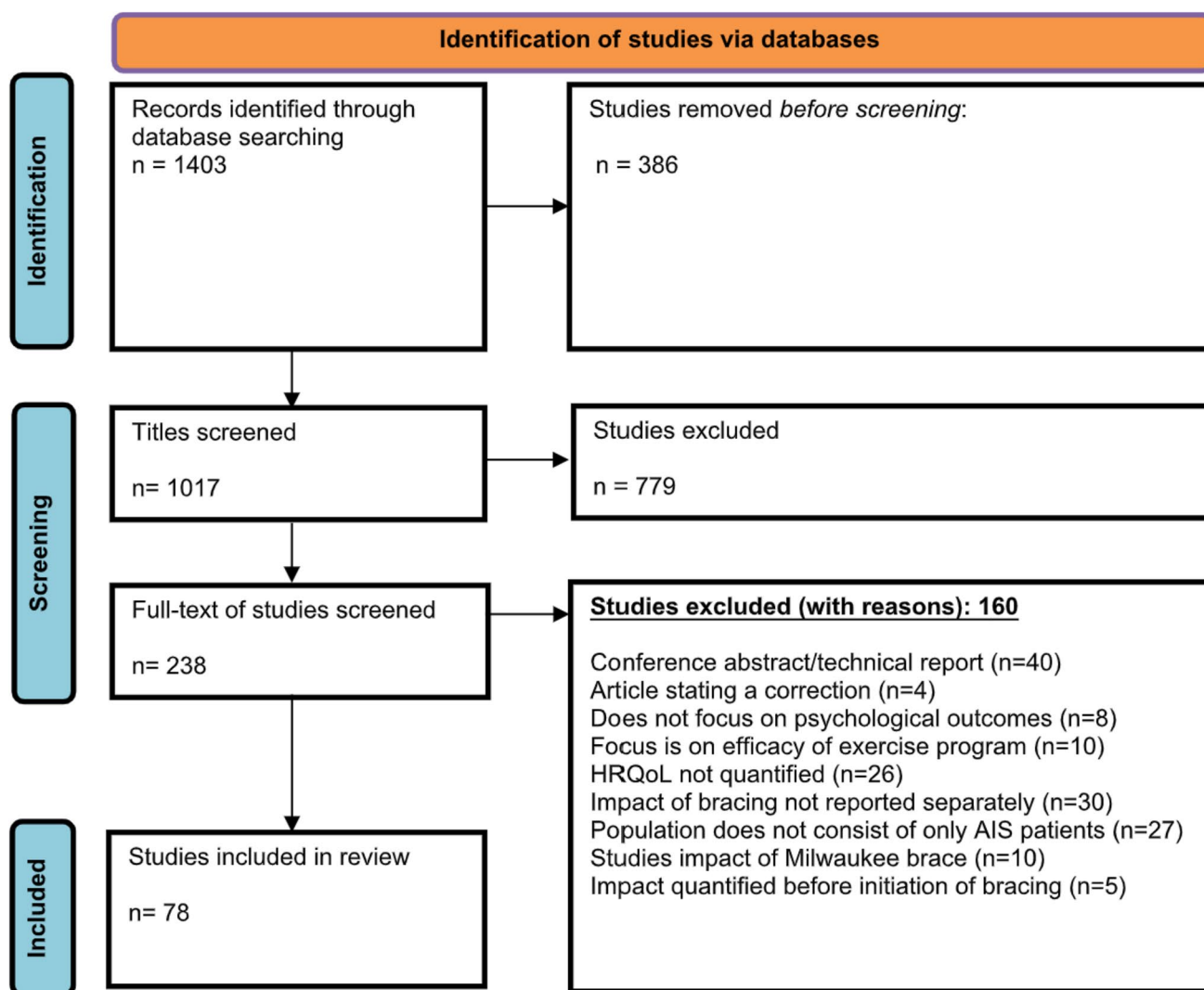


Fig. 1 PRISMA flow diagram

Narrative synthesis

The narrative synthesis reveals a wide landscape of mixed evidence across psychological, social and physical domains. These are summarised in Table 3 and expanded in the discussion. While consistent patterns emerge regarding treatment burden and long-term adaptation for patients, the overall findings are characterised by varying levels of evidential strength.

Quantitative data

This summarises the quantitative data available for different subgroups of papers reporting SRS-22. The Minimal Clinically Important Difference for the SRS-22 is 0.4 [103]. There was insufficient data to summarise other questionnaires. These analyses are from studies of different designs,

in different countries of non-homogenous AIS populations with similar but not identical brace indications, different brace types and different brace regimes. The analysis is mainly aimed to represent the variability in the data across studies.

SRS-22 during bracing

SRS-22 scores during bracing were available for 51 unique groups of AIS patients in 41 studies, comprising a total of 4193 participants. Data collection timepoints ranged from three to 32 months post-bracing initiation. The pooled mean SRS-22 score across all braced participants at all timepoints was 4.05 (95% CI: 3.96, 4.13). The forest plot in Fig. 2 of all included studies demonstrates generally high scores (above 4) for the SRS-22 during bracing, with some variation. As expected, the heterogeneity was extremely high (98%)

Table 1 Characteristics of included studies: study design, bracing criteria, brace types, questionnaires and risk of bias

First Author, Year	Study design (in relation to questionnaire)	Country	Sample size	Age at initiation of bracing	Initial mean ($\pm$ SD) Cobb angle (degrees)	SRS bracing criteria met?	Brace type	Questionnaires used	Risk of bias tool	Risk of bias total	Intervention category*	Included in quantitative summaries?
Liu, 2021 [22]	Cross-sectional	China	70	10 to 18 years	Not specified	Yes	Not specified	BrQ SRS-22	MINORS	Low	AF	Yes
Park, 2021 [23]	Cross-sectional	Korea	120	Not specified	Not specified	No	Not specified	ISYQOL SRS-22	MINORS	Low	AF	
Feise, 2005 [24]	Cross-sectional	Canada	84	Not specified	Not specified	No	Not specified	SQOL QLPSD	MINORS	Moderate	AF	
Pham, 2008 [25]	Cross-sectional	France	108	Not specified	FTB 30.5 $\pm$ 3.1 Part-time brace 29.2 $\pm$ 4.3	No	Cheneau	QLPSD VAS of pain	MINORS	Moderate	AF	
Asada, 2021 [26]	Cross-sectional	Japan	69	Not specified	30.9 $\pm$ 5.2	No	Boston	J-BSSQ-Brace SRS-22 (Japanese)	MINORS	Moderate	AF	Yes
Babae, 2014 [27]	Cross-sectional	Iran	60	Not specified	30.77 $\pm$ 8.5	No	TLSO	SRS-22	MINORS	Moderate	AF	Yes
Mitchell, 2022 [28]	Cross-sectional	USA	200	Not specified	33 $\pm$ 15	No	Not specified	PROMIS CAT SRS-22r (validity)	Cochrane RoB2	Moderate	AF	Yes
Mousavi, 2010 [29]	Cross-sectional	Iran	84	12–18 years	Not specified	No	Not specified	SRS-22 SF 36	MINORS	Moderate	AF	
Guy, 2020 [30]	Randomised controlled trial	Canada	94	$\geq$ 10 years	28 $\pm$ 7	Yes	TLSO	SRS-22r	MINORS	Low	BT	Yes
Chua, 2023 [31]	Cross-sectional	Singapore	730	8–21 years	Not specified	No	Not specified	SRS-22r EQ-5D-5L EQ-VAS	MINORS	Moderate	BT	Yes
Lin, 2020 [32]	Randomised controlled trial	Hong Kong	23	10–14 years	Pressure-adjustable orthosis—26.2 $\pm$ 4.7 Conventional orthosis—27.4 $\pm$ 5.6	Yes	Not specified	SRS-22 (Chinese) BrQ (Chinese)	MINORS	Moderate	BT	Yes
Lin, 2021 [33]	Randomised controlled trial	Hong Kong	30	10–14 years	3D-printed orthosis—31.8 $\pm$ 5.8 Conventional orthosis—29.3 $\pm$ 4.0	Yes	3D printed and conventional	SRS-22r (Chinese) TAPS (Chinese) BrQ (Chinese)	MINORS	Moderate	BT	Yes
Misterska, 2012 [34]	Cross-sectional	Poland	126	10–17 years	Not specified	Yes	Cheneau	BSSQ-deformity (Polish) BSSQ-Brace (Polish)	Cochrane RoB2	Moderate	BT	
Zheng, 2018 [35]	Randomised controlled trial	Hong Kong	53	10–17 years	Bracing group—28.00 $\pm$ 3.60	Yes	TLSO	SRS-22	Cochrane RoB2	Moderate	BT	Yes

Table 1 (continued)

First Author, Year	Study design (in relation to questionnaire)	Country	Sample size	Age at initiation of bracing	Initial mean ($\pm$ SD) Cobb angle (degrees)	SRS bracing criteria met?	Brace type	Questionnaires used	Risk of bias tool	Risk of bias total	Intervention category*	Included in quantitative summaries?
Asada, 2024 [36]	Retrospective Cohort	Japan	122	10–14 years	Not specified	No	Boston	SRS-22r SJ-27 (females) Japanese BSSQ-Brace	Cochrane RoB2	Moderate	BT	Yes
Kastrinis, 2023 [37]	Cross-sectional	Greece	47	Not specified	30.8 $\pm$ 9.25	No	SLC Clinic-asymmetric	SRS-22 (Greek) BSSQ-Brace (Greek) BSSQ-Deformity (Greek)	MINORS	Moderate	BT	
Danielsson, 2013 [38]	Cross-sectional	Sweden	193	Not specified	Not specified	No	Not specified	SRS-22r (Swedish) SF-36 (Swedish)	MINORS	Moderate	BT	Yes
Simony, 2014 [39]	Cross-sectional	Denmark	207	Not specified	Not specified	No	Boston	SRS-22 (Danish) SF-36 (Danish)	COSMIN	Moderate	BT	
Maria, 2021 [40]	Cross-sectional	Italy	56	11–16 years	27.2 $\pm$ 6.24	No	Sforzesco and Cheneau	BSSQ-Brace ISYQoL	MINORS	Low	LNC	
Aulisa, 2010 [41]	Cross-sectional	Poland	108	Not specified	PASB—28.0 $\pm$ 1.2 Lyon—35.2 $\pm$ 1.6	No	PASB and Lyon	SRS-22 Brace Questionnaire	MINORS	Low	LNC	Yes
Bunge, 2007 [42]	Cross-sectional	Netherlands	109	Not specified	Brace treated—24 $\pm$ 8.5 Brace and surgery—34 $\pm$ 12.6	No	Boston and Triac	BSSQ-brace SRS-22 (Dutch) VAS	Cochrane RoB2	Low	LNC	Yes
Wang, 2022 [43]	Cross-sectional	China	17	10–17 years	Not specified	Yes	Boston and Cheneau	SRS-22 (Chinese) GCQ (Chinese) ODI (Chinese)	MINORS	Low	LNC	Yes
Misterska, 2014 [44]	Cross-sectional	Poland	41	10–17 years	Not specified	Yes	Cheneau	SAQ (SAQ-pl) (Polish) SAQ-pl parent form (Polish)	COSMIN	Low	LNC	
Andersen, 2006 [45]	Cross-sectional	Denmark	583	9–21 years	Not specified	No	Boston	SF-36	COSMIN	Low	LNC	
Bae, 2023 [46]	Cross-sectional	Korea	372	8–19 years	Not specified	No	Not specified	BDI (Korean) EQ-5D (Korean) EQ-VAS (Korean) OBC-Y (Korean)	COSMIN	Low	LNC	

Table 1 (continued)

First Author, Year	Study design (in relation to questionnaire)	Country	Sample size	Age at initiation of bracing	Initial mean ($\pm$ SD) Cobb angle (degrees)	SRS bracing criteria met?	Brace type	Questionnaires used	Risk of bias tool	Risk of bias total	Intervention category*	Included in quantitative summaries?
Cantelea, 2023 [47]	Prospective cohort	Italy	68	10–16 years	Good compliance— 27 ± 9 Poor compliance— 23 ± 7	Yes	Cheneau, Agostini, Lionese and Sforzesco	SRS-22r (Italian)	COSMIN	Low	LNC	Yes
Glowacki, 2013 [48]	Prospective cohort	Poland	36	10–17 years	Not specified	Yes	Cheneau	SAQ-Polish (SAQ-pl) STAIC-trait	MINORS	Moderate	LNC	
Glowacki, 2013 [49]	Prospective cohort	Poland	36	10–17 years	Not specified	Yes	Cheneau	SDQ-25 (Polish)	MINORS	Moderate	LNC	
Li, 2024 [50]	Cross-sectional	China	103	10–14 years	3DPB group— 32.6 ± 4.85 TLSO group— 31.7 ± 5.62	Yes	3D Printed and TLSO	SRS-22 (Chinese) EQ-5D-3L (Chinese) EQ-VAS (Chinese)	MINORS	Moderate	LNC	Yes
Piantoni, 2018 [51]	Cross-sectional	Argentina	43	10–18 years	Not specified	No	TLSO	Brace Questionnaire (Spanish)	MINORS	Moderate	LNC	
Misterska, 2012 [52]	Prospective Cohort	Poland	36	10–17 years	Not specified	Yes	Cheneau	TAPS SRS-22 BSSQ-deformity	MINORS	Moderate	LNC	Yes
Misterska, 2011 [53]	Cross-sectional	Sweden	64	10–17 years	Urban— 26.8 ± 5.6 Rural— 28.2 ± 4.3	No	Cheneau	BSSQ-brace BSSQ-deformity	MINORS	Moderate	LNC	
Cheung, 2007 [54]	Cross-sectional	China	92	10–20 years	28.8 ± 9.97	No	Boston and TriaC	SRS-22	COSMIN	Moderate	LNC	
Vasiliadis, 2006 [55]	Cross-sectional	Greece	36	Not specified	Not specified	No	Boston	BrQ	COSMIN	Moderate	LNC	
Weigert, 2006 [56]	Cross-sectional	Denmark	118	12–20 years	Braced— 33.2 ± 6.6 Braced and surgery— 44.9 ± 10.5	No	Boston	SRS-24 (Danish) BrQ	COSMIN	Moderate	LNC	
Farshad, 2022 [57]	Cross-sectional	Switzerland	20	10–20 years	Brace treated— 35 ± 18.6	No	Not specified	EQ-5D-5L SF-36 SRS24 ODI PGWBI	COSMIN	Moderate	LNC	
Simony, 2015 [58]	Cross-sectional	Denmark	170	Not Specified	Not specified	No	Boston	SF-36 (Danish) SRS-22R (Danish) EQ-5D (Danish)	MINORS	Moderate	LNC	Yes

Table 1 (continued)

First Author, Year	Study design (in relation to questionnaire)	Country	Sample size	Age at initiation of bracing	Initial mean ($\pm$ SD) Cobb angle (degrees)	SRS bracing criteria met?	Brace type	Questionnaires used	Risk of bias tool	Risk of bias total	Intervention category*	Included in quantitative summaries?
Baymurat, 2023 [59]	Cross-sectional	Turkey	51	10–17 years	31.3 $\pm$ 3.6	Yes	TLSO	SRS-22	MINORS	Moderate	LNC	
Brigham, 2016 [60]	Cross-sectional	USA	39	10–16 years	29 degree	No	Boston	Scoliosis Compliance Questionnaire SRS-22 Qualitative exploratory questions on motivations and obstacles	COSMIN	Moderate	LNC	Yes
Chan, 2014 [61]	Prospective Cohort	Hong Kong	42	10 years or above	29.4 $\pm$ 3.9	Yes	TLSO	TAPS (Chinese) BrQ (Chinese) SRS-22 (Chinese)	COSMIN	Moderate	LNC	
Danielsson, 2010 [62]	Cross-sectional	Sweden	77	11–15 years	Brace treated: 30.5 $\pm$ 3.2	Yes	Boston	SF-36 SRS-22 (Swedish)	COSMIN	Moderate	LNC	Yes
Rivett, 2009 [63]	Cross-sectional	South Africa	31	13–16 years	Not specified	No	Cheneau	BrQ	MINORS	Moderate	LNC	
Cheung, 2020 [64]	Retrospective Cohort	Hong Kong	586	10–14 years	31 $\pm$ 4	Yes	Boston	SRS-22r	COSMIN	Moderate	LNC	
Chopraa, 2022 [65]	Prospective Cohort	Scotland	24	10–16 years	Bracing group—27 $\pm$ 4	No	TLSO	SRS-22	MINORS	Moderate	LNC	Yes
Alsiddiky, 2024 [66]	Cross-sectional	Saudi Arabia	52	$\geq$ 10 years	(median and range) Boston—29.7 (24.5–34.1) Chéneau—31.5 (27.5–37.9)	Yes	Boston and Cheneau	SRS-22r BSSQ-Brace	MINORS	High	LNC	
Salah, 2023 [67]	Cross-sectional	Tunisia	48	10–18 years	Not specified	No	Cheneau	QLPSD SRS-22 VAS EVA-QOL	MINORS	High	LNC	
Torén, 2022 [68]	Cross-sectional	Sweden	307	10–18 years	Not specified	Yes	TLSO and Hyperextension night-time	SRS-22r and EQ-5D index (Swedish)	MINORS	High	LNC	Yes
Lertudomphonwanit, 2021 [69]	Cross-sectional	Thailand	40	10–18 years	33 $\pm$ 8.4	No	Boston	Brace Questionnaire (Thai)	MINORS	High	LNC	
Ersen, 2016 [70]	Cross-sectional	Turkey	68	10–15 years	SpineCor Group—35.1 $\pm$ 8.2 Rigid brace Group—33.5 $\pm$ 7.5	Yes	TLSO	SRS-22 (Turkish)	MINORS	High	LNC	Yes

Table 1 (continued)

First Author, Year	Study design (in relation to questionnaire)	Country	Sample size	Age at initiation of bracing	Initial mean ($\pm$ SD) Cobb angle (degrees)	SRS bracing criteria met?	Brace type	Questionnaires used	Risk of bias tool	Risk of bias total	Intervention category*	Included in quantitative summaries?
Pellios, 2016 [71]	Cross-sectional	Greece	77	> 10 years	28.25 $\pm$ 8.7	No	Boston	SF-36 Greek) ODI (Greek)	COSMIN	High	LNC	
Lim, 2024 [72]	Prospective Cohort	Singapore	24	10–12 years	33.6 $\pm$ 7.5	Yes	TLSO	SRS-22	MINORS	Low	OT	
Cheung, 2016 [73]	Prospective cohort	Hong Kong	227	10–18 years	25.0 $\pm$ 11.4	No	Not specified	EQ-5D-SL SRS-22	COSMIN	Low	OT	Yes
Cheung, 2018 [74]	Cross-sectional	Hong Kong	652	10–18 years	27.7 $\pm$ 10.7	No	Not specified	EQ-5D-SL SRS-22	MINORS	Low	OT	
Yuan, 2024 [75]	Randomised controlled trial	China	94	Not Specified	Not specified	No	Not specified	SF-36	COSMIN	Low	OT	
Caronni, 2019 [76]	Cross-sectional	Italy	742	10–18 years	Not specified	No	Not specified	SRS-22 ISYQOL	MINORS	Moderate	OT	Yes
Asada, 2023 [77]	Retrospective cohort	Japan	41	12.4 $\pm$ 1.4 years	30.0 $\pm$ 4.4	No	Boston	J-BSSQ (Brace)	MINORS	Moderate	OT	
Peeters, 2023 [78]	Cross-sectional	Netherlands	80	12–18 years	38 $\pm$ 12	No	Boston and Cheneau	SRS-22r (Dutch) BrQ (Dutch)	MINORS	Moderate	OT	Yes
Peuchot, 2024 [79]	Prospective Cohort	France	38	15 $\pm$ 2.7 years	Not specified	No	PASB, Cheneau or Art brace followed by corset by corset Collette	SRS-22 BrQ	MINORS	Moderate	OT	
Forest, 2023 [80]	Cross-sectional	Canada	111	Not specified	28 $\pm$ 12	No	Not Specified	SRS-22 ISYQOL-F SF-12	MINORS	Moderate	OT	Yes
Misterska, 2019 [81]	Cross-sectional	Poland	35	12–16 years	27.8 $\pm$ 7.4	Yes	Cheneau	BSSQ-Brace BSSQ-Deformity	MINORS	Moderate	OT	
Misterska, 2019 [82]	Cross-sectional	Poland	71	10–14 years	SpineCor (SC) group: 25.9 $\pm$ 5.0 Cheneau (Ch) group: 26.9 $\pm$ 5.6	Yes	SpineCor and Cheneau	SRS-22 BrQ SAQ PODCI	MINORS	Moderate	OT	Yes
Giir, 2017 [83]	Prospective cohort	Turkey	28	10–17 years	31.1 $\pm$ 8.3 (thoracic) 29.7 $\pm$ 9.6 (lumbar)	No	TLSO	BrQ BSSQ-Brace SRS-22	MINORS	Moderate	OT	Yes
Cheung, 2019 [84]	Retrospective cohort	Hong Kong	265	10.0 to 15.9 years	31.0 $\pm$ 3.8	Yes	Boston	SRS-22	MINORS	Moderate	OT	Yes

Table 1 (continued)

First Author, Year	Study design (in relation to questionnaire)	Country	Sample size	Age at initiation of bracing	Initial mean ($\pm$ SD) Cobb angle (degrees)	SRS bracing criteria met?	Brace type	Questionnaires used	Risk of bias tool	Risk of bias total	Intervention category*	Included in quantitative summaries?
Lim, 2017 [85]	Cross-sectional	Korea	103	10–16 years	28.7 $\pm$ 5.9	No	Boston	Brace questionnaire SRS-22	MINORS	Moderate	OT	
Carreon, 2011 [86]	Cross-sectional	USA	1802	Not specified	55.8 $\pm$ 13.7	No	Not specified	SAQ SRS-22	MINORS	Moderate	OT	
Wong, 2022 [87]	Randomised controlled trial	Hong Kong	38	10–15 years	11.7 $\pm$ 6 1.0	Yes	TLSO	BrQ SRS-22	COSMIN	Moderate	OT	
Weiss, 2007 [88]	Prospective cohort	Germany	63	Not specified	43.7 $\pm$ 16.1	No	Cheneau, Cheneau light, Boston, Cheneau-Boston-Wiesbaden, Wilmington, SpineCor	BSSQ	COSMIN	Moderate	OT	
Duarte, 2022 [89]	Retrospective cohort	Canada	80	10–15 years	33.4 $\pm$ 6.8 (thoracic) 29.8 $\pm$ 8.6 (lumbar)	No	Not specified	SRS-22	COSMIN	Moderate	OT	
Fang, 2022 [90]	Retrospective cohort	China	192	10–16 years	Not specified	Yes	Cheneau	SRS-22 EQ-5D-5L EQ-VAS	COSMIN	Moderate	OT	Yes
Vasiladis, 2006 [91]	Cross-sectional	Greece	28	Not specified	23.2 (thoracic) 21.2 (lumbar)	No	Boston	BrQ	COSMIN	Moderate	OT	
Zhang, 2011 [92]	Prospective cohort	China	65	Not specified	Group 1 (30.14 $\pm$ 5.18) Group 2 (45.28 $\pm$ 2.80)	No	Not specified	Children's Self-Concept Scale	MINORS	Moderate	OT	
Sanders, 2003 [93]	Cross-sectional	USA	138	10–21 years	Not specified	No	Not specified	WRVAS Subjective "deformity noted" or "deformity not noted"	MINORS	Moderate	OT	
Ólafsson, 1999 [94]	Prospective cohort	Sweden	54	8.4–17.3 years	33	No	Boston	"I think I am" questionnaire	Cochrane RoB2	Moderate	OT	
Xu, 2015 [95]	Prospective cohort	China	86	Not specified	30.5 $\pm$ 1.5	No	Not specified	BSSQ-Brace BSSQ-deformity SRS-22	Cochrane RoB2	Moderate	OT	Yes
Zhang, 2021 [92]	Cross-sectional	China	217	Not specified	32.7 $\pm$ 5.35	No	Cheneau	BrQ (Chinese) SRS-22 (Chinese)	COSMIN	Moderate	OT	Yes

Table 1 (continued)

First Author, Year	Study design (in relation to questionnaire)	Country	Sample size	Age at initiation of bracing	Initial mean ($\pm$ SD) Cobb angle (degrees)	SRS bracing criteria met?	Brace type	Questionnaires used	Risk of bias tool	Risk of bias total	Intervention category*	Included in quantitative summaries?
Kyrkousis, 2024 [96]	Randomised controlled trial	Greece	80	Not specified	Control: 34.65 $\pm$ 3.08 intervention: 33.43 $\pm$ 5.98	No	Not specified	SRS-22	COSMIN	Moderate	OT	Yes
Liu, 2021 [97]	Cross-sectional	China	138	10 to 18 years	Not specified	Yes	Not specified	ISYQOL SRS-22	COSMIN	Moderate	OT	Yes
Zhao, 2007 [98]	Cross-sectional	China	86	10–18 years	35.6	No	Not specified	SRS-22	COSMIN	High	OT	

References correspond to the reference list in the main manuscript

*LNC = longitudinal with no comparator; BT = comparison between two or more different brace types; OT = comparison between bracing and other treatments, i.e. surgery, exercise or no treatment i.e. observation; AF = comparison between bracing and another factor (including regimens and compliance)

BDI beck depression inventory, *BrQ* brace questionnaire, *BSSQ-Brace* Bad Sobernheim Stress Questionnaire—Brace, *BSSQ-Deformity* Bad Sobernheim Stress Questionnaire—Deformity, *EQ-5D/EQ-5D-3L/EQ-5D-5L* EuroQol 5-Dimensions, *EQ-VAS* EuroQol Visual Analogue Scale, *EVA-QOL* echelle visuelle analogue—quality of life, *ISYQOL/ISYQOL-F* Italian Spine Youth Quality of Life, *OBQ-Y* Oxford back care—youth, *ODI* Oswestry disability index, *PGWBI* psychological general well-being index, *PODCI* paediatric outcomes data collection instrument, *PROMIS* CAT patient-reported outcomes measurement information system, *QLPSD* quality of life profile for spinal deformities, *SAQ* scoliosis appearance questionnaire, *SDQ-25* strengths and difficulties questionnaire 25 items, *SF-12* short form-12 health survey, *SF-36* short form-36 health survey, *SI-27* spinal appearance questionnaire 27-item version, *SQLI* scoliosis quality of life index, *SRS-22/SRS-22r* Scoliosis Research Society-22, *SRS-24* Scoliosis Research Society-24, *STAIC* state-trait anxiety inventory for children, *TAPS* trunk appearance perception scale, *VAS* visual analogue scale, *WRVAS* Walter Reed Visual Assessment Scale

reflecting the variation in study types and data collection methods.

SRS-22 during bracing and no treatment

Eight studies had data for both bracing and observation, including 2374 participants in total (observation = 1196, bracing = 1178). As shown in the forest plot in Fig. 3, five studies favoured observation, two favoured bracing and one reported no difference. Most patients in the observation group had smaller curves than the braced patients. The two studies favour bracing, one evaluated patients at least 11 years after bracing [62] and one had 41% of patients in a night-time brace with some observed patients awaiting surgery [68]. However, the pooled effect size remained small at -0.11 (95% CI: $-0.21, 0.00$), with high statistical heterogeneity (83%) across these eight studies. This is less than the SRS-22 MCID.

Within the domains of the SRS-22, the function and self-image domain more often favoured no treatment over bracing in the studies, pain and mental health had high variability between studies and satisfaction usually favoured bracing in the studies.

SRS-22 during bracing and surgery

Data for both bracing and surgery were available from seven studies. Indications for surgery were either not given or were variable (curves > 45 – 50 degrees). As illustrated in Fig. 4 forest plot, results were highly inconsistent. Three studies favoured bracing, three favoured surgery and one demonstrated no difference.

Within the different domains of the SRS-22, pain, self-image and mental health domains showed variation in the small number of studies that had data ($n = 4$). Function was more likely to favour surgery but satisfaction more often favoured bracing.

Risk of bias

Overall, most studies fall into the moderate risk of bias category (Table 4). Of the observational studies assessed by the MINORS tool, 69% were moderate [25–27, 31, 34, 36, 41, 42, 45, 46, 48, 49, 52, 54, 56–59, 61–63, 66–68, 70–72, 74, 82, 88, 89, 93, 94, 102] with 21% low risk [40, 47, 50, 53, 64, 65, 69, 79, 84, 90] and only 10% high risk [43, 44, 51, 55, 60]. Similarly, of the validation studies assessed by the COSMIN tool, most studies showed moderate risk (68%) [23, 28, 29, 37, 39, 73, 76–78, 80, 81, 86, 91, 97, 98], with 23% low risk [22, 24, 85, 92, 95] and 9% high risk [38, 83]. The RCTs had slightly less risk of bias; 86% had moderate risk of bias [30, 32, 33, 75, 87, 96], with just one study

achieving low risk [35] and none with high risk. Full details of assessments can be found in Online Appendix 3.

Discussion

Psychological well-being

Psychological well-being was influenced by brace regimen and compliance with treatment. Multiple studies suggested longer brace wear (especially when over 20 h per day) was associated with negative emotional impacts and increased stress [40, 47, 63, 69]. Cantele et al. suggested that higher compliance with prescribed duration was associated with better long-term psychological adjustment and self-image, despite worse stress in the short term [47]. This suggests that the psychological and emotional costs incurred during bracing may be offset by the psychological benefits post-bracing.

Overall, surgical treatment is generally associated with better psychological well-being. Multiple studies suggested that AIS patients treated surgically scored substantially higher on the self-image and mental health sub-scales of SF-36 or SRS-22 than those treated with a brace, but these differences were not statistically significant [42, 45, 58].

Similarly, studies found that bracing was associated with higher stress and decreased satisfaction when compared with observation as well as lower mental health when compared with exercise alone [54, 67]. Differences were less evident in studies that assessed psychological well-being years after bracing [57, 62], as mental health was more affected by other factors such as self-perception and coping mechanisms [62]. Overall, psychological outcomes appear most sensitive during active bracing, with the negative impact diminishing over time, after bracing completion.

This trend persists among braced cohorts, revealing a pattern of transient, early-stage decline in the psychological well-being of patients with AIS. Multiple studies reported increased anxiety and worry or reduced mood in the period following brace initiation [23, 43, 48, 49, 59, 78, 79, 102]. However, longitudinal cohorts revealed partial recovery or stabilisation of these effects [26, 48, 55, 64, 83, 91]. The moderators repeatedly reported across studies with a negative effect on psychological scores were longer prescribed bracing duration, linked to higher early distress and pre-existing body-image vulnerability. Parental and patient support, as well as clinical parameters with higher curve correction or improvement, led to better psychological scores [49, 55, 59, 61, 79]. However, factors such as heterogeneity in questionnaires used, lack of pre-bracing outcomes and variable timings of measurements limit causal inference.

When comparing brace types, a small number of studies reported directional effects in favour of less rigid brace designs on psychological domains of questionnaires, while

other studies reported no discernible difference between brace types [27, 32, 41, 50, 66]. One study suggested worse psychosocial outcomes associated with a bulkier, older brace [82]. The hypothesised causal pathway was consistent across studies: lesser brace-related stress led to better body image which in turn resulted in improved mood and good psychological adjustment. However, confidence in this causal relationship is reduced by the heterogeneity in reported PROMs, the timing of outcome assessments, a predominance of observational study designs and lack of adjustment for bracing adherence or curve severity.

Social functioning and peer relationships

Stricter regimens and extended daily brace wear were found to increase the degree of social burden [40, 47, 63]. Adolescents wearing a brace for longer per day reported greater disruption to social activities and feelings of embarrassment [40, 63, 69], though one study reported good adherence being associated with improved peer interactions [69]. The results from long-term follow-up studies indicated social disadvantages diminished after the end of bracing [47, 63, 71].

Evidence directly comparing social outcomes for bracing versus surgery is limited, but their individual impacts have been explored [45, 58, 68]. Studies reported that surgical patients found it easier to resume social life after recovery, where bracing negatively affected peer relationships, social confidence and physical activity [45, 56, 58]. A higher degree of social discomfort and stigmatisation were seen even after decades post brace treatment when compared with surgery [58].

It was difficult to isolate the direct effects of bracing compared with conservative treatment (e.g. exercise or observation) on social functioning and peer relationships, as these were embedded within broader mental well-being or self-image measures [54, 62, 68]. However, braced adolescents were more likely to report that they encountered difficulties socially when compared with observation groups [54, 67, 68].

Studies of braced groups report social self-consciousness, temporary reduction in participation in sports and peer awkwardness and teasing; again mainly in the initial months of treatment with gradual social adaptation over time [27, 48, 60, 78, 97]. Positive moderators included reduced brace visibility, opportunities for wardrobe adaptation and peer support, whereas restrictive school activities amplify negative effects [22, 59, 79, 83].

More comfortable and less visible brace designs are often assumed to help reduce barriers and social embarrassment, but no statistically significant differences were reported in data collected using the SRS-22 questionnaire and the EQ-5D-3L tool [32, 50, 70]. Most inferences rely only on the self-image domain of SRS-22 or qualitative commentary

Table 2 Outcome measures used within the included studies

Outcome	No of studies <i>n</i> = 78
SRS-22	53 (68%)
BrQ	17 (22%)
BSSQ-brace	14 (18%)
SF-36	9 (12%)
EQ-5D-5L	6 (8%)
BSSQ-deformity	6 (8%)
Other*	26 (33%)

*Outcomes used five times or less—including iterations of reporting outcomes such as the EQ-5D-3L and SF-12

by the author [27, 66]. Studies suggested a plausible, but sparsely evidenced pattern that less conspicuous brace types cause less social disruptions [32, 33, 50, 70, 82].

Physical wellbeing and body perception

Regimens where bracing was prescribed for longer per day was associated with greater negative effects including discomfort, impaired mobility and pain during treatment, as well as worse body image [40, 47, 63]. Studies with extended follow-up, during bracing and after treatment, indicated that better adherence led to less curve progression, however, differences in long-term functional outcomes are inconsistent, with no significant differences in overall SRS-22 or SF-36 scores in the long term [47, 63, 71]. It is inconclusive whether the temporary negative effects of stricter regimens are outweighed by long-term benefits to QoL and mental health [40, 63, 71].

Surgery results in greater curve correction than bracing, leading to higher satisfaction with appearance and better body image, as well as physical functioning [42, 45, 58]. Ongoing dissatisfaction with body image was frequently reported by adolescents treated with a brace along with discomfort and pain [58, 68]. Both groups reported comparable long-term functional outcome scores and low disability [56, 58]. Scores from two studies measuring HRQoL at 10 years and 25 years after treatment indicated that surgically treated patients had a superior long-term HRQoL [45, 58].

Bracing was consistently linked with more functional limitations and reduced physical comfort than conservative treatment or observation [54, 67, 68]. However, other factors such as curve severity and progression appear to determine physical outcomes [57, 62]. Braced adolescents felt more negative about their bodies during bracing, with these differences becoming less apparent in adulthood [57, 62, 67, 68]. Long-term comparisons at 42 years post-treatment indicated

spinal curve progression, not bracing history, was the dominant determining factor of outcomes [57].

In cross-sectional and cohort studies of brace-treated AIS patients, the majority reported a modest level of transient discomfort, activity-related limitations and skin irritation early in bracing [34, 44, 55, 61]. Many adolescents adapted over months with infrequent objective negative long-term effects on physical well-being [27, 78, 88]. Cosmetic dissatisfaction and body perception are more strongly associated with lower psychosocial outcomes [23, 27, 83] with functional scores remaining moderate [23, 48, 51, 97]. Longitudinal data indicate that these outcomes can typically stabilise or improve over time, particularly when bracing achieves good curve reduction [55, 61, 84, 88], although some studies find persistent issues with body image [51, 74]. Important moderators across studies were brace design, sporting restrictions and the number of hours of daily brace use, however, these measurements were predominantly self-reported and variable [23, 27, 61, 88, 91]. Inconsistent measuring instruments, variable timepoints and heterogeneous scoring conventions make direct comparison across studies difficult.

Where comparing brace types, a number of studies report that braces providing adjustability, improved fit or those that were less bulky were associated with fewer activity limitations, less discomfort and better self-image compared with rigid brace designs [30, 32, 33, 50, 70, 82]. However, not all studies found significant differences in the sub-scales of pain or function and there were a number that did not adjust for adherence [27, 50, 66]. This is noteworthy as adherence may be a key mediator between comfort and physical outcomes.

Comparisons with other literature

Previous reviews focussed on biomechanical or radiographic outcomes, rather than QoL [104, 105]. Our results are consistent with published qualitative reports describing social withdrawal, embarrassment or discomfort [106–109], but also highlight positive trends suggesting that the psychosocial effects of bracing are largely reversible and less severe than previously implied. This is consistent with other chronic paediatric conditions, where emotional disruption is attenuated by coping mechanisms developed by patients [110, 111]. These findings collectively suggest that psychological burden of bracing is real, but may be modifiable through supportive interventions and patient-centred brace designs.

Strengths and limitations

To our knowledge, this systematic review is the most comprehensive synthesis to date exploring the QoL and psychological consequences of bracing in AIS. Three established risk of bias assessment tools were applied to ensure that the quality of each study was assessed validly. The convergent

Table 3 Key themes from narrative synthesis

Domain	Key findings	Strength of evidence/statistical significance
Psychological well-being	Brace treatment associated with increased anxiety and reduced mood during the period of active treatment, noted particularly in case of longer daily wear	Often reported but is largely based on observational data; QoL differences reflect some significant findings
	Improved psychological well-being over time during bracing and post-bracing	Improvement over time suggested by longitudinal data, however, limited in strength owing to variability in measures
	Bracing adherence associated with improved outcomes, potentially offsetting short-term distress	Mixed evidence with some studies showing significant QoL improvements and others suggesting no overall difference
	Surgical treatment associated with better emotional functioning and self-image, these differences often not statistically significant	Occasional significant differences with findings often non-significant
	Perception of self and coping mechanisms contributed more to psychological outcomes than treatment type	Evidence is suggestive with limited causal inference
	Higher social burden and reduced participation in cases of stricter bracing regimens	Consistent trend across findings, however, data were largely self-reported or observational
Social Functioning and Peer Relationships	Early bracing period is associated with more prominent social difficulties with a certain degree of adaptation over time	Longitudinal studies demonstrate some post-bracing recovery
	More social challenges reported by braced patients when compared with exercise or observation	Evidence present, however, often embedded within broader measures of QoL
	Surgically treated patients may find it easier to socially reintegrate after the end of treatment	Significant advantages in social reintegration reported in some studies but often embedded within larger QoL measures, limiting direct comparison
	Social burden may be reduced when the brace is less bulky or less visible	No statistically significant differences in EQ-5D-3L or SRS-22; evidence is mostly qualitative
	During the active treatment period, longer duration of brace wear associated with activity limitation, pain and greater discomfort	Consistent findings, but mostly self-reported and short-term
Physical Well-being and Body Perception	Curve progression reduced in cases of high adherence	Findings clinically significant and supported across studies
	Surgery associated with higher satisfaction and better body image, however, long-term functional outcomes are similar	Results are mixed as there are some important HRQoL advantages even though the functional outcomes are often comparable
	Physical impact is more influenced by the progression or curve severity rather than the type of treatment	Suggestive evidence which is supported by long-term data
	Improvements in brace design linked to better self-image and more comfort	Mixed evidence with some positive trends, but are not consistently statistically significant

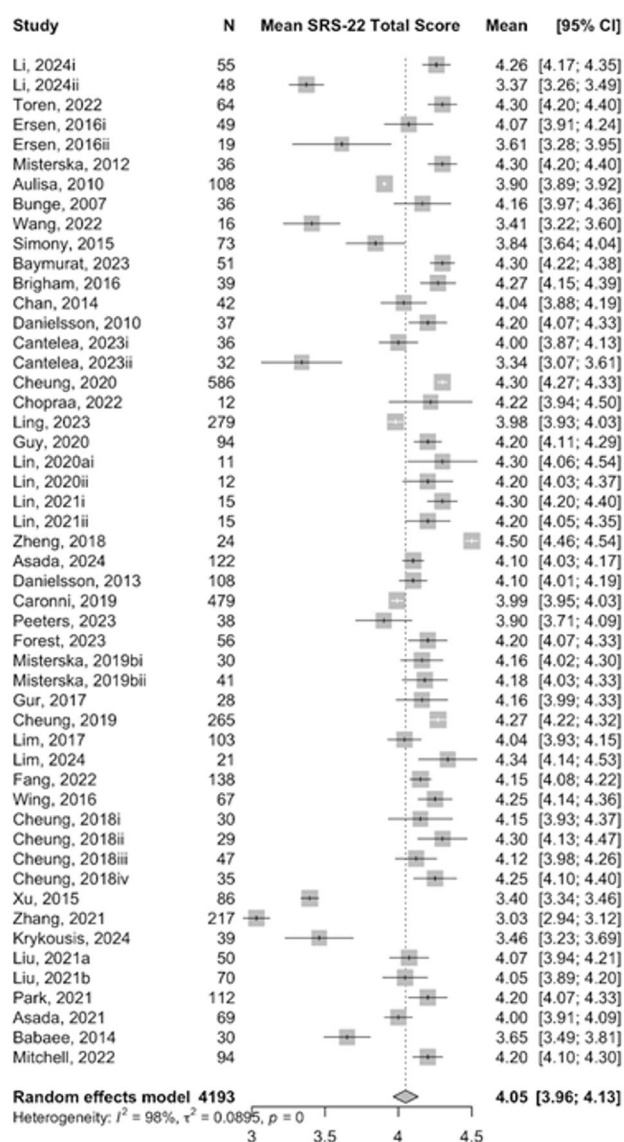


Fig. 2 A forest plot of SRS-22 scores during bracing

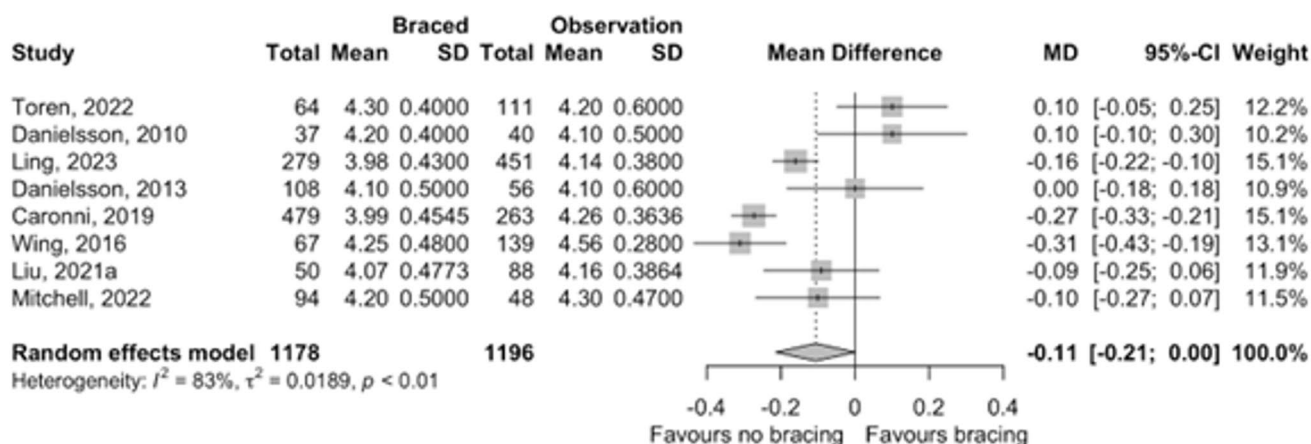


Fig. 3 A forest plot of SRS-22 scores during bracing and observation

results and evidence across independent continents and studies increase confidence in the overall findings.

However, conclusions are restricted by high heterogeneity across studies, reflecting variability in measurement tools, brace type, adherence definitions and cultural context. Methodological shortcomings in the included studies frequently involved small sample sizes, cross-sectional and retrospective designs, inconsistent or short follow-up periods, and limited adjustment for confounding variables.

Unfortunately, only a small number of studies had sufficient data to be included in the pooled analysis. It is also noted that compliance could be considered a determinant of treatment effectiveness, and therefore may independently affect psychosocial outcomes, limiting the interpretation of pooled results. Additionally, the lack of data regarding total treatment duration limits these conclusions as large variation in bracing duration is likely a key factor in perceived quality of life. There is also the difficulty of isolating the impact of the brace from the psychological burden of the scoliosis itself. These challenges are further compounded by significant heterogeneity in the pooled analysis, with data collection timepoints ranging from 3 to 32 months post-initiation, limiting a standardised assessment of adjustment over time.

Implications for clinicians and future research

While bracing is clearly effective, this treatment does carry a discernible psychosocial cost that should be considered, with braced patients showing a lower self-image and satisfaction scores compared to surgical treatment in the narrative synthesis. While the pooled SRS-22, as well as the function and self-image domains, showed a small, non-statistically significant difference between braced and non-braced adolescents (SRS-22 in favour of bracing, and function and self-image in favour of surgery), the reductions may still be of clinical importance as the heterogeneity between studies is

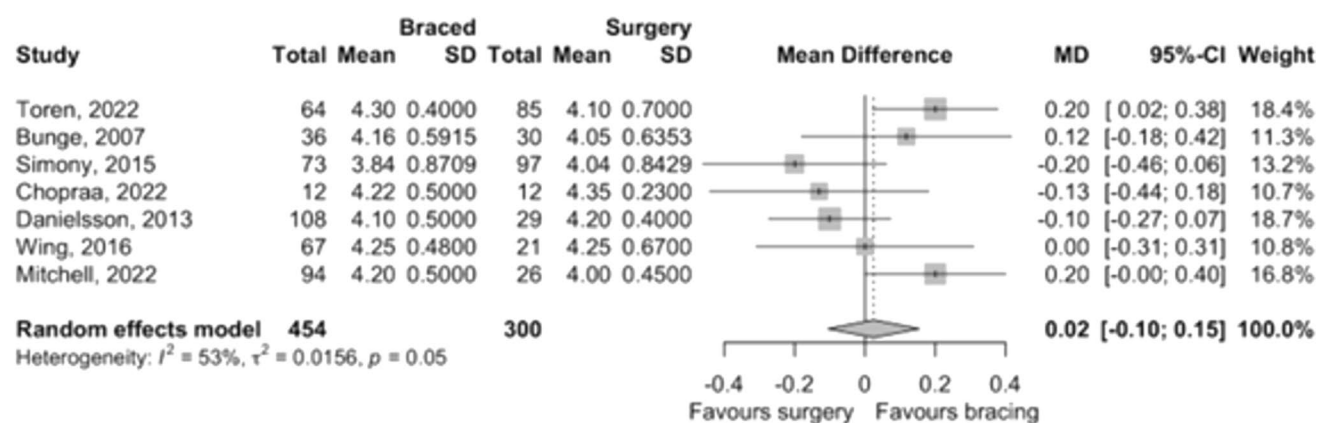


Fig. 4 A forest plot of SRS-22 scores during bracing and surgery

Table 4 Risk of bias summary

Tool	Risk of bias level	Number of studies
MINORS (<i>n</i> = 49)	Low	10 (21%)
	Moderate	34 (69%)
	High	5 (10%)
Cochrane risk of bias (<i>n</i> = 7)	Low	1 (14%)
	Moderate	6 (86%)
	High	0 (0%)
COSMIN (<i>n</i> = 22)	Low	5 (23%)
	Moderate	15 (68%)
	High	2 (9%)

high and may be making a difference. Adolescence represents a period of critical development and is characterised by heightened body awareness and identity formation [112, 113].

The visible and restrictive nature of bracing poses significant psychosocial challenges that must be addressed to ensure treatment adherence and success in clinical practice. Clinicians should integrate structured psychological support, such as family education, counselling and peer mentorship, into brace management to improve compliance and mental well-being. Early psychological distress should be recognised as a modifiable risk factor. Using validated PROMs at the start of bracing and during follow-up can identify those at risk for social withdrawal, anxiety or depression. Clinicians must consider that more intensive bracing imposes greater psychosocial and physical burden during brace treatment, yet appears to result in superior long-term physical and mental outcomes due to better curve stabilisation.

Multidisciplinary interventions involving orthotists, physiotherapists and psychologists are important. The physical and functional limitations imposed by bracing may intensify the psychological burden on patients. Consequently, designers should focus on improving comfort and aesthetics to

mitigate the overall impact on patient well-being. Bulkier, more rigid orthoses and longer prescribed use were also associated with lower self-image, higher stress and greater social withdrawal. Conversely, customised or modern brace designs focussing on aesthetics improved self-image and comfort without compromising effectiveness. Future research should focus on prospective, multicentre studies with larger cohorts and validated assessment tools. These studies should incorporate objective measures regarding treatment duration, bracing indications and compliance with both the prescribed bracing regimen and SRS criteria.

Conclusion

This review suggests that brace treatment has a modest, negative effect on mental health and QoL, particularly impacting social functioning, self-image and emotional well-being. Long-term data suggest some improvement after brace treatment completion. Overall, healthcare providers should be aware that bracing in AIS carries a discernible psychosocial cost which may affect brace adherence, potentially impacting on overall bracing success. We need to understand the association between the effects of brace treatment on all aspects of QoL and how this impacts brace adherence and success. We need to consider psychological support in those who need it, and aesthetic, more comfortable brace designs. Future research should prioritise large-scale, prospective multicentre studies that combine validated QoL measures with objective compliance monitoring and adherence to the SRS bracing criteria.

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Author contributions All authors contributed to the study's protocol development and design. Material preparation and data acquisition

were performed by AN, MB, CH, NT and KR, with NT also conducting the statistical analysis. AN prepared the initial manuscript draft. All authors critically reviewed and approved the final manuscript. Project supervision was provided by KR, LS and AC.

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Material availability All data in this review are publicly available.

Declarations

Conflict of interest AC and NT are co-applicants on the ongoing NIHR funded BASIS trial (NIHR131081) from which LS and KR have received funding for salaries. The other authors have no relevant financial or nonfinancial interests to disclose. The authors did not receive support from any organisation for the submitted work.

Ethical approval Not applicable, all data are publicly available.

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